

# Final Project Topic Proposal

## Team 1

**Group Members:** Venkatesh Nagarjuna, Dhruv Rai, Sadheen Hossain

### **Research topic: Food Deserts in the United States**

This project investigates the geographic, demographic, and socioeconomic factors contributing to food deserts across the United States—areas where access to affordable, fresh, and nutritious food is severely limited. By analyzing the USDA's Food Access Research Atlas and Food Environment Atlas, we aim to uncover key indicators correlated with limited food access and generate data-driven policy recommendations to guide federal intervention strategies.

The following are the initial set of tentative SMART questions that we have set forth:

1. Where are food deserts most geographically concentrated across the U.S., and how do these concentrations differ between urban and rural census tracts?
2. What demographics and socioeconomic factors are linked to food deserts?
3. How are food deserts related to health issues like obesity and diabetes?
4. Can we develop a predictive model that accurately identifies high-risk areas likely to be or become food deserts based on social and economic indicators?
5. What national policies could reduce food deserts based on our findings?

**Dataset link:** <https://www.ers.usda.gov/data-products/food-access-research-atlas/download-the-data>

**GitHub Link:** [https://github.com/Asbetos/data\\_mining\\_final](https://github.com/Asbetos/data_mining_final)

**Modeling methods:** Classification and clustering